

Benchmarking Computational Methods for RNA Methylation Detection from Nanopore Direct RNA Sequencing

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Supplementary Results

A Cross-species m6A benchmark

In the human HEK293T m6A benchmark, the raw candidate spaces differed substantially among the four tools. TandemMod, m6Anet, CHEUI, and SWARM covered 855,533, 203,667, 103,358, and 15,023 candidate sites, respectively, whereas the common candidate universe shared by all four tools contained only 9,447 sites. Under the main high-confidence reference setting, 516 positive sites and 8,398 pseudo-negative sites were finally included in the evaluation. Fixed-threshold results showed that SWARM achieved the highest F1 score, whereas m6Anet and CHEUI exhibited relatively high precision. In contrast, the positive call set of TandemMod was markedly restricted under the high-confidence site-level calling rule.

In the mouse mESC m6A benchmark, the four tools shared 10,143 common candidate sites. After label construction based on the high-confidence positive set and the loose exclusion set, 8,428 DRACH sites were finally included, consisting of 277 reference-positive sites and 8,151 pseudo-negative sites, with a positive proportion of 3.29%. Under fixed thresholds, SWARM achieved the highest recall and F1 score, m6Anet achieved the highest precision, CHEUI showed a conservative pattern with high precision and high specificity, and TandemMod retained a certain level of recall but had relatively low precision.

In the *Arabidopsis* VIRc m6A benchmark, 40,083 sites remained in the exact common candidate universe after ambiguous sites were excluded, including 549 reference-positive sites and 39,534 pseudo-negative sites, with a positive proportion of 1.37%. Fixed-threshold evaluation showed that m6Anet achieved the highest F1 score. SWARM achieved the highest recall but had relatively lower precision; CHEUI showed intermediate performance; and TandemMod produced a markedly limited number of positive calls under this high-confidence calling criterion.

Overall, SWARM showed strong fixed-threshold performance and ranking ability in the human and mouse m6A benchmarks, particularly in terms of recall, F1 score, and continuous modification-level concordance. m6Anet maintained stable ranking performance across all three species and achieved the highest F1 score in the *A. thaliana* dataset. CHEUI generally behaved as a more conservative caller, whereas the performance of TandemMod was more strongly affected by the site-level calling rule and candidate-site aggregation strategy.

SUPPLEMENTARY TABLE S1. FIXED-THRESHOLD PERFORMANCE IN CROSS-SPECIES M6A BENCHMARK

Dataset	Tool	TP	FP	FN	TN	Precision	Recall	Specificity	F1
Human HEK293T	TandemMod	10	79	506	8319	0.112	0.019	0.991	0.033
Human HEK293T	m6Anet	217	21	299	8377	0.912	0.421	0.997	0.576
Human HEK293T	CHEUI	193	17	323	8381	0.919	0.374	0.998	0.532
Human HEK293T	SWARM	420	62	96	8336	0.871	0.814	0.993	0.842
Mouse mESCs	TandemMod	187	512	90	7639	0.268	0.675	0.937	0.482
Mouse mESCs	m6Anet	144	45	133	8106	0.762	0.520	0.994	0.618
Mouse mESCs	CHEUI	120	40	157	8111	0.750	0.433	0.995	0.549
Mouse mESCs	SWARM	236	218	41	7933	0.520	0.852	0.973	0.646
<i>Arabidopsis</i> VIRc	TandemMod	15	37	534	39497	0.288	0.027	0.999	0.050
<i>Arabidopsis</i> VIRc	m6Anet	251	509	298	39025	0.330	0.457	0.987	0.383
<i>Arabidopsis</i> VIRc	CHEUI	177	400	372	39134	0.307	0.322	0.990	0.314
<i>Arabidopsis</i> VIRc	SWARM	382	1215	167	38319	0.239	0.696	0.969	0.356

SUPPLEMENTARY TABLE S2. AUROC AND AUPRC IN CROSS-SPECIES M6A BENCHMARK

Dataset	Tool	AUROC	AUPRC
Human HEK293T	TandemMod	0.640	0.085

Human HEK293T	m6Anet	0.967	0.788
Human HEK293T	CHEUI	0.861	0.636
Human HEK293T	SWARM	0.973	0.896
Mouse mESCs	TandemMod	0.877	0.344
Mouse mESCs	m6Anet	0.972	0.689
Mouse mESCs	CHEUI	0.842	0.533
Mouse mESCs	SWARM	0.980	0.824
<i>Arabidopsis</i> VIRc	TandemMod	0.775	0.042
<i>Arabidopsis</i> VIRc	m6Anet	0.976	0.309
<i>Arabidopsis</i> VIRc	CHEUI	0.918	0.249
<i>Arabidopsis</i> VIRc	SWARM	0.931	0.309

In human HEK293T and mouse mESCs, we further evaluated the concordance between tool-derived continuous outputs and GLORI_ratio. In the human HEK293T dataset, SWARM showed the highest concordance with GLORI_ratio, with Pearson correlation, Spearman correlation, and R^2 values of 0.895, 0.833, and 0.800, respectively, and RMSE and MAE values of 0.125 and 0.091, respectively. m6Anet ranked second when mod_ratio was used, CHEUI showed moderate concordance, whereas TandemMod exhibited relatively weaker continuous-level agreement.

In the mouse mESC complete-case analysis, 119 GLORI reference sites had valid continuous predictions from all four tools: TandemMod, m6Anet, CHEUI, and SWARM. SWARM performed best across both correlation and error metrics, indicating that its stoichiometry output could better reflect the modification levels estimated by GLORI. The mod_ratio output of m6Anet also showed good numerical concordance and outperformed probability_modified when used as a continuous modification-level measure. CHEUI stoichiometry captured part of the variation in modification levels, but its overall concordance was slightly weaker than that of m6Anet. TandemMod p95_ratio showed a certain degree of correlation with GLORI_ratio, but its numerical error was relatively higher.

SUPPLEMENTARY TABLE S3. CONTINUOUS AGREEMENT WITH GLORI_RATIO IN M6A DATASETS

Dataset	Tool	Pearson	Spearman	R2	RMSE	MAE
Human HEK293T	TandemMod	0.163	0.235	0.026	0.434	0.352
Human HEK293T	m6Anet	0.819	0.766	0.670	0.153	0.114
Human HEK293T	CHEUI	0.630	0.552	0.397	0.234	0.157
Human HEK293T	SWARM	0.895	0.833	0.800	0.125	0.091
Mouse mESCs	TandemMod	0.6845	0.6302	0.4685	0.2201	0.1629
Mouse mESCs	m6Anet	0.7647	0.6991	0.5847	0.1713	0.1352
Mouse mESCs	CHEUI	0.7503	0.6272	0.5630	0.1846	0.1402
Mouse mESCs	SWARM	0.8963	0.8223	0.8034	0.1215	0.0951

B Continuous modification-level agreement and human m5C cross-modification evaluation

In the cross-modification evaluation of human m5C, the fixed-threshold performance of the three tools was markedly weaker than that observed in the human m6A task. Under the official thresholds, SWARM achieved the highest Precision and F1 score, with values of 0.002277 and 0.004520, respectively, while its Recall was 0.300. CHEUI achieved a Precision of 0.001123 and an F1 score of 0.002209, but its Recall was only 0.0667. TandemMod showed a Precision of 0.000110, an F1 score of 0.000230, and a Recall of 0.1126. Although relaxed thresholds increased the Recall of some tools, with CHEUI, TandemMod, and SWARM reaching Recall values of 0.6833, 0.3841, and 0.425, respectively, Precision and F1 did not improve accordingly. For example, under the relaxed threshold, CHEUI achieved a Precision of only 0.000130 and an F1 score of 0.000260.

Unlike the m6A task, the continuous outputs of TandemMod, CHEUI, and SWARM in the human m5C evaluation showed Pearson and Spearman correlations close to zero with the GSE93749 HeLa-siCTRL final modification levels, and most R^2 values were negative. These results indicate that, in the m5C task, the continuous scores produced by current tools are more appropriately interpreted as site-level confidence or detection scores, rather than as quantitatively comparable modification proportions across platforms. The m5C results suggest reduced cross-modification generalizability and further indicate more complex inconsistencies among external reference systems, sequencing technologies, and tool-specific output scales.

SUPPLEMENTARY TABLE S4. HUMAN m5C FIXED-THRESHOLD AND RELAXED-THRESHOLD EVALUATION

Tool	Threshold type	Threshold rule	Precision	Recall	F1
TandemMod	Official-like	p95 > 10 and p95_ratio >= 0.2	0.000110	0.1126	0.000230
TandemMod	Relaxed	p95_ratio >= 0.2	0.000045	0.3841	0.000091
CHEUI	Official	probability >= 0.9999	0.001123	0.0667	0.002209
CHEUI	Relaxed	probability >= 0.7	0.000130	0.6833	0.000260
SWARM	Official	probability >= 0.9972 and stoichiometry > 0.1	0.002277	0.3000	0.004520
SWARM	Relaxed	probability >= 0.7	0.000440	0.4250	0.000890

C Downsampling-based robustness of m6A calling

The *Arabidopsis* downsampling analysis showed that, as the read fraction was gradually reduced from full depth to 20%, most tools exhibited an overall decline in both reference-positive recovery and the number of positive calls. Under the full-depth condition, SWARM recovered the largest number of reference-positive sites, with 382 out of 549 sites detected, corresponding to a reference-hit recall of 0.696. m6Anet ranked second, recovering 251 out of 549 reference-positive sites, with a reference-hit recall of 0.457. CHEUI recovered 177 out of 549 sites, corresponding to a reference-hit recall of 0.322, whereas TandemMod recovered only 15 out of 549 sites, with a reference-hit recall of 0.027.

At the mildly downsampled 80% read fraction, the reference-hit recall values of m6Anet and SWARM remained close to those observed under the full-depth condition. m6Anet recovered 250 reference-positive sites at the 80% read fraction, with a reference-hit recall of 0.455 and a full-depth TP-site retention rate of 96.0%. SWARM recovered 375 reference-positive sites, with a reference-hit recall of 0.683 and a full-depth TP-site retention rate of 97.1%. CHEUI also retained a large proportion of full-depth positive sites at the 80% read fraction, with a reference-hit recall of 0.317 and a full-depth TP-site retention rate of 93.8%. For TandemMod, because the number of detectable reference-positive sites under the full-depth condition was already very small, a slight fluctuation was observed at the 80% read fraction, where 17 reference-positive sites were recovered, corresponding to a reference-hit recall of 0.031 and a full-depth TP-site retention rate of 86.7%. This change is more likely to reflect sampling variation under low detection counts and should not be interpreted as a genuine performance improvement.

SUPPLEMENTARY TABLE S5. DOWNSAMPLING SUMMARY FOR M6A REFERENCE-HIT RECOVERY

Tool	TandemMod	m6Anet	CHEUI	SWARM
Full recall	0.027	0.457	0.322	0.696
80% recall	0.031	0.455	0.317	0.683
60% recall	0.018	0.444	0.288	0.672
40% recall	0.016	0.421	0.286	0.643
20% recall	0.011	0.311	0.193	0.481

SUPPLEMENTARY TABLE S6. POSITIVE-CALL COMPOSITION UNDER SELECTED READ FRACTIONS

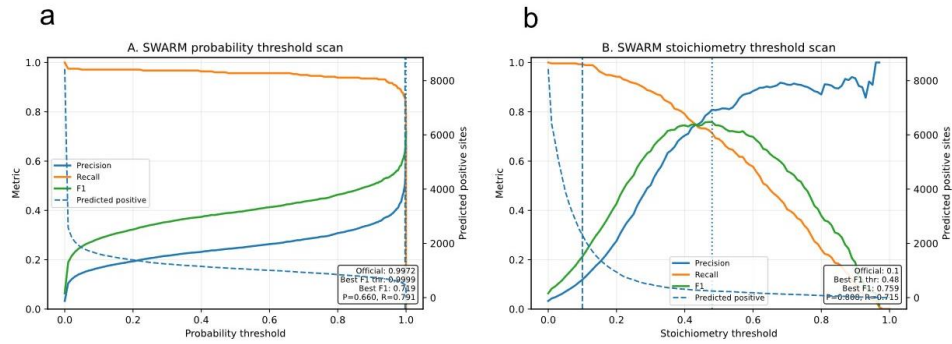
	Reads fraction	m6Anet	SWARM
positive calls	100%	760	1597
	80%	675	1423
	60%	589	1204
	40%	456	899
	20%	250	448
non-reference calls	100%	509	1215
	80%	425	1048
	60%	345	835
	40%	225	546
	20%	79	184

D SWARM threshold sensitivity analysis

The SWARM threshold scan showed that modification probability and stoichiometry had different effects on the Precision–Recall trade-off. In the probability-only threshold scan, increasing the probability cutoff reduced the number of predicted positive sites and improved Precision, whereas Recall declined in the high-threshold range. Without applying an additional stoichiometry filter, the best probability-only F1 score was 0.719, obtained at a cutoff close to 0.9999, with a Precision of 0.660 and a Recall of 0.791.

The stoichiometry-only scan showed a more evident Precision–Recall trade-off. The best F1 score reached 0.759 at a stoichiometry threshold of approximately 0.48, with a Precision of 0.808 and a Recall of 0.715. The joint threshold scan further showed that combining probability and stoichiometry thresholds could better balance Precision and Recall. In the mouse common-site benchmark, the best F1 score was observed around probability = 0.9899 and stoichiometry = 0.35, where Precision, Recall, and F1 reached 0.790, 0.801, and 0.796, respectively.

In contrast, the official fixed criteria, defined as probability ≥ 0.9972 and stoichiometry > 0.1 , were more recall-oriented, with a Recall of 0.852, a Precision of 0.520, and an F1 score of 0.646. These results suggest that SWARM performance can be improved by task-specific joint threshold calibration rather than relying only on the official fixed setting.



Supplementary Fig. S1 | Single-threshold scanning of SWARM filtering attributes in the mouse common-site benchmark.

(a) Probability-only threshold scan. **(b)** Stoichiometry-only threshold scan. Solid curves indicate Precision, Recall, and F1 score across different thresholds, while the dashed curve represents the number of predicted positive sites. Vertical dashed lines mark the official threshold and the F1-maximizing threshold. The probability-only scan achieved a best F1 score of 0.719 at a cutoff close to 0.9999, whereas the stoichiometry-only scan reached a higher best F1 score of 0.759 at a threshold of approximately 0.48.

E Comparison of SWARM and m6Anet positive call sets

The comparison between SWARM and m6Anet positive call sets revealed an asymmetric overlap pattern. SWARM predicted 454 positive sites, among which 236 were reference-positive sites, corresponding to a Precision of 0.520, a Recall of 0.852, and an F1 score of 0.646. In comparison, m6Anet predicted 189 positive sites, among which 144 were reference-positive sites, with a Precision of 0.762, a Recall of 0.520, and an F1 score of 0.618.

The intersection of the two tools contained 171 sites, including 141 reference-positive sites. This strategy achieved the highest Precision among all evaluated strategies, reaching 0.825, but its Recall decreased to 0.509. The union of SWARM and m6Anet only slightly improved Recall compared with SWARM alone, increasing it to 0.863, but Precision decreased to 0.506 and F1 did not exceed that of SWARM alone.

These results indicate that most high-confidence m6Anet-positive sites were already included in the SWARM call set. Therefore, if the goal is to obtain a more conservative and high-confidence set of m6A sites, the intersection of SWARM and m6Anet may serve as a useful filtering strategy.

F Effect of SegPore-derived eventalign on m6Anet performance

The SegPore-derived m6Anet analysis showed that the upstream eventalign source affected both the candidate space and score calibration of m6Anet. In the full DRACH candidate space, the Nanopolish-derived workflow covered 102,606 evaluable sites, of which 2,309 overlapped with the high-confidence truth set. In contrast, the SegPore-derived workflow covered 44,046 evaluable sites, with 1,024 sites overlapping with the high-confidence truth set.

The Nanopolish-derived m6Anet workflow achieved an AUROC of 0.967 and an AUPRC of 0.661, both higher than those of the SegPore-derived workflow, which achieved an AUROC of 0.899 and an AUPRC of 0.287. Under the fixed threshold, the Nanopolish-derived workflow showed higher Precision, F1, and Specificity, whereas the SegPore-derived workflow showed higher Recall but lower Precision. This pattern suggests that the SegPore-derived score distribution was shifted toward higher values, and the default m6Anet cutoff could not be directly transferred to the SegPore-derived input.

We further compared the two workflows on GGACT motif-associated common candidate sites. The traditional m6Anet workflow identified 5,576 GGACT genome sites, whereas the SegPore-derived m6Anet workflow identified 2,462 GGACT

genome sites, with 1,775 sites commonly covered by both workflows. On these shared sites, the probability_modified scores showed relatively high consistency, with Pearson and Spearman correlation coefficients of 0.726 and 0.851, respectively. This indicates that SegPore-derived eventalign still preserved ranking signals similar to those of the Nanopolish-derived workflow within the shared candidate space.

Among 866 GGACT common evaluation sites, the traditional m6Anet and SegPore-derived m6Anet workflows achieved AUROC values of 0.969 and 0.934, respectively, and AUPRC values of 0.937 and 0.885, respectively. Under the fixed threshold of probability_modified ≥ 0.9 , the traditional m6Anet workflow achieved a Precision of 0.945, a Recall of 0.763, and an F1 score of 0.844. In contrast, the SegPore-derived workflow achieved a much higher Recall of 0.987, but its Precision decreased to 0.541 and its F1 score was 0.699.

After increasing the SegPore-derived m6Anet threshold to 0.97, Precision improved to 0.787 and F1 increased to 0.829, while Recall remained relatively high at 0.877. These results suggest that SegPore-derived eventalign retains modification-related signals, but SegPore-specific threshold calibration or further model adaptation is required for high-confidence calling.

G Sensitivity of SegPore-derived m6Anet to min_segment_count

The min_segment_count sensitivity analysis showed that increasing the read-support threshold substantially reduced the output space of SegPore-derived m6Anet, while its effect on overall ranking performance was relatively moderate. As min_segment_count increased from 20 to 50, the number of evaluable sites decreased from 44,080 to 21,894, and the number of high-confidence predicted sites with probability_modified ≥ 0.9 decreased from 8,014 to 3,680.

In the fixed-threshold evaluation, Precision increased from 0.158 to 0.177, F1 increased from 0.261 to 0.285, and AUPRC increased from 0.286 to 0.323, whereas Recall only slightly decreased from 0.753 to 0.738. AUROC remained largely stable at approximately 0.90. These results indicate that more stringent read-support filtering can remove part of the low-confidence output space and moderately improve prediction reliability, but it cannot fully resolve the score calibration differences associated with SegPore-derived input.

H Computational resource profiling

The computational resource assessment showed substantial differences in runtime and memory requirements among the evaluated tools. TandemMod and CHEUI required longer total runtimes, and their computational costs increased further in the m5C task. For TandemMod, the total runtime increased from 310:09:17 in the m6A dataset to 718:57:25 in the m5C dataset, mainly due to the increased time required for m5C preprocessing and prediction.

CHEUI also showed long total runtimes in both the m6A and m5C datasets, and its host memory peak reached 264.56 GB during the m5C inference stage. In contrast, SWARM had the shortest runtime among the evaluated tools, with total runtimes of 6:23:44 and 21:43:37 in the m6A and m5C tasks, respectively.

These results indicate that practical tool selection should not be based only on detection performance. Data scale, hardware availability, supported modification types, and computational cost should also be considered when selecting DRS-based RNA methylation callers for large-scale analyses.

SUPPLEMENTARY TABLE S7. MIN_SEGMENT_COUNT SENSITIVITY FOR SEGPORE-DERIVED M6ANET

Workflow	SegPore-derived m6Anet	SegPore-derived m6Anet	SegPore-derived m6Anet	SegPore-derived m6Anet
min_segment_count	20	30	40	50
Callable sites	44,080	33,232	26,704	21,894
Positive eval sites	1,024	820	674	568

Negative eval sites	36,732	27,791	22,439	18,461
Positive calls at cutoff 0.9	4,887	3,637	2,930	2,368
TP	771	609	500	419
FP	4,116	3,028	2,430	1,949
FN	253	211	174	149
TN	32,616	24,763	20,009	16,512
Precision	0.158	0.167	0.171	0.177
Recall	0.753	0.743	0.742	0.738
Specificity	0.888	0.891	0.892	0.894
F1	0.261	0.273	0.277	0.285
AUROC	0.899	0.897	0.896	0.899
AUPR	0.286	0.306	0.306	0.323

Supplementary Notes

A Overview of the benchmarking strategy

This study addresses the practical requirements for using RNA modification detection tools in nanopore direct RNA sequencing settings and establishes a supplementary evaluation framework covering four aspects: cross-species performance, cross-modification applicability, sequencing-depth variation, and tool-parameter sensitivity. While the main text presents the overall benchmarking framework and core findings, this supplementary material further describes the data sources, reference-set construction, candidate-site definition, tool-output harmonization, threshold selection, and additional sensitivity analysis procedures, thereby providing a clearer and more reproducible basis for the performance comparisons reported in the main text.

TABLE S8. SUMMARY OF RNA MODIFICATION CALLERS INCLUDED IN THIS STUDY.

Tool	Version/Commit	Main supported task in this study
m6Anet	v2.1.0	m6A calling
TandemMod	v1.1.0	m6A and m5C calling
CHEUI	Git commit 7b422f7808a3c2ffff56a9ead33a199824753b4e	m6A and m5C calling
SWARM	Git commit 741332e30651a22fbc032b09c47f3fd1ec71d53	m6A and m5C calling

B Dataset collection and benchmark tasks

Nanopore direct RNA sequencing datasets were selected according to the evaluation tasks, public data availability, and the availability of external references. The main cross-species m6A benchmark included a mouse mESCs Mettl3 wild-type direct RNA-seq dataset, two human HEK293T-WT replicates, and an *Arabidopsis thaliana* VIRc direct RNA-seq dataset. The cross-modification m5C evaluation used human HeLa-related data, together with publicly available bisulfite-seq or RNA-BisSeq evidence to construct external reference sets.

In the main m6A benchmark, all four tools were evaluated on the mouse, human, and *A. thaliana* datasets. For the cross-modification m5C evaluation, m6Anet was not included because it is primarily designed for m6A modification modeling; therefore, only TandemMod, CHEUI, and SWARM were assessed.

C m6A callable universe, label construction, and reference-set construction

In the m6A evaluation, to reduce the influence of differences in callable regions across tools on direct comparisons, the main analysis was performed within an exact-site common universe shared by all four tools. Outputs from all tools were converted into species-specific genome-level scorebed files and restricted to the corresponding m6A candidate-site background.

For the mouse and human datasets, sites supported by at least two independent sources were defined as the high-confidence positive set, whereas sites supported by any source were defined as the loose exclusion set. Sites falling within the high-confidence positive set were labeled as positive. Sites located near the loose exclusion set but not overlapping the high-confidence positive set were treated as ambiguous sites and excluded from the evaluation. The remaining shared candidate sites were defined as pseudo-negatives. It should be emphasized that pseudo-negatives were not experimentally validated true negatives, but low-evidence candidate sites used for relative comparison among tools. Given a common candidate universe (U), a high-confidence positive set (P), and a loose reference set (L), the final labels were defined as follows:

$$positive = U \cap P$$

$$ambiguous = U \cap buffer(L)$$

$$pseudo - negative = U - positive - ambiguous$$

Continuous agreement analysis was performed only when an external quantitative reference matched to the corresponding direct RNA-seq dataset was available and provided site-level modification ratios. The mouse m6A continuous reference was constructed from DRACH sites quantified in both mESC GLORI replicates, and the human m6A continuous reference was constructed from DRACH sites quantified in both HEK293T-WT GLORI replicates. For each site, the mean GLORI_ratio across the two replicates was used as the reference modification level. Tool outputs from the mouse and human datasets were harmonized to mm39 and GRCh38 genome-level coordinates, respectively, and matched to the corresponding GLORI continuous references by exact genomic coordinates. To ensure comparability across tools, the main continuous agreement analysis followed a complete-case strategy, retaining only sites with valid values in GLORI and in all included tools. Per-tool available-site results were reported as supplementary analyses. Since no matched site-level continuous modification-ratio reference was available for the *Arabidopsis thaliana* direct RNA-seq dataset, GLORI-like continuous agreement analysis was not performed for the *A. thaliana* benchmark.

D m5C callable universe, label construction, and reference-set construction

To further evaluate the applicability of RNA modification callers across modification types, a human m5C cross-modification evaluation was performed in addition to the main m6A benchmark. Because m6Anet is primarily designed for m6A modeling, this analysis included only TandemMod, CHEUI, and SWARM, which are officially described as applicable to m5C detection.

The human m5C reference sets were constructed from publicly available HeLa bisulfite-seq or RNA-BisSeq datasets, including GSE122260, GSE140995, and GSE93749. Data from different sources were converted into genome-level BED format and lifted over from hg19 to GRCh38 coordinates. Sites supported by at least two sources were defined as the high-confidence m5C positive set, whereas the union of all three sources was used as the loose exclusion set.

Given the substantial differences in candidate-site coverage among tools in the m5C task, fixed-threshold evaluation was performed within a per-tool callable universe rather than a common universe. Specifically, pseudo-negative sites were constructed within the sites for which each tool produced valid scores, after excluding the loose exclusion set and nearby ambiguous regions. This design avoided unstable estimates caused by an overly small common universe and preserved tool-specific callable-region differences.

In addition to fixed-threshold evaluation, GSE93749 HeLa-siCTRL Final Level was used as the continuous m5C reference to assess the agreement between tool-derived continuous scores and external modification levels. For this analysis, TandemMod used p95_ratio, whereas CHEUI and SWARM used stoichiometry as continuous scores.

E Tool-output harmonization, threshold selection, and score-field definition

All raw nanopore direct RNA-seq FAST5 files were first basecalled with Guppy to generate FASTQ files. Reads were then aligned to the corresponding species-specific transcriptomes using minimap2 to generate BAM files. Tool-specific signal-level preprocessing and alignment were performed according to each official workflow, including Nanopolish-based event alignment or FAST5/single-read signal conversion and resquiggle-based signal realignment where applicable. After prediction, site-level outputs from each tool were extracted and converted into species-specific genome-level scorebed files. All downstream evaluations were performed using genome-level coordinates. For tools with potential 0-based/1-based coordinate differences or transcript-to-genome conversion differences, the most appropriate coordinate representation was determined by reference overlap and shift-scan analyses.

For threshold-dependent metrics, this study prioritized cutoffs recommended in tool documentation, original publications, or official workflows. For outputs without a unified site-level calling threshold, we adopted rules representing high-confidence modification proportions or high-confidence read support. To ensure comparability among tool outputs, fixed-threshold scoring fields and continuous modification-ratio fields were selected according to the meaning of each tool’s output.

For ROC and PR curve analyses, TandemMod used p0.95/total as the continuous score, m6Anet used site-level probability_modified, and CHEUI and SWARM used site-level modification probability. For continuous modification-level agreement analysis, the field most closely corresponding to the modification ratio was selected for each tool: TandemMod used p0.95/total, m6Anet used mod_ratio from data.site_proba.csv, and CHEUI and SWARM used site-level modification stoichiometry. The scoring fields and fixed-threshold definitions for each tool are summarized in Table S9.

TABLE S9. SCORE FIELDS AND FIXED POSITIVE RULES USED FOR M6A EVALUATION.

Tool	Continuous score for ROC/PR	Continuous score for modification-level concordance	Fixed positive rule for m6A
TandemMod	p0.95 / total	p0.95 / total	p0.95 > 10 and p0.95 / total ≥ 0.2
m6Anet	probability_modified	mod_ratio	probability_modified ≥ 0.9
CHEUI	site-level probability	site-level probability	probability ≥ 0.9999
SWARM	modification probability	modification stoichiometry	modification probability ≥ 0.9972 and modification stoichiometry > 0.1

For the human m5C evaluation, m6Anet was excluded from this analysis because it is primarily designed for m6A modification modeling. The fixed-threshold evaluation of m5C was performed using both the official thresholds and relatively relaxed/unified thresholds to examine the effects of threshold selection on precision, recall, and F1 score. The detailed threshold definitions are provided in Table S10.

TABLE S10. OFFICIAL AND RELAXED THRESHOLD DEFINITIONS USED FOR THE HUMAN M5C EVALUATION.

Tool	Official threshold	Relaxed / unified threshold
TandemMod	p95 > 10 and p95_ratio ≥ 0.2	p95_ratio ≥ 0.2
CHEUI	probability ≥ 0.9999	probability ≥ 0.7
SWARM	modification probability ≥ 0.9972 and modification stoichiometry > 0.1	modification probability ≥ 0.7

F Evaluation metrics

For discrete reference sets, site-level continuous scores were converted into positive or negative calls according to the recommended or high-confidence cutoffs defined for each tool. Based on these binary calls, true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN) were calculated. Fixed-threshold metrics included precision, recall, specificity, accuracy, and F1 score. Precision reflects the proportion of positive calls supported by the reference set, recall reflects the recovery of reference-positive sites, and F1 score summarizes the balance between precision and recall.

In addition to fixed-threshold evaluation, thresholds were scanned across the full score range to generate receiver operating characteristic (ROC) and precision–recall (PR) curves, from which AUROC and AUPRC were calculated. Because RNA modification sites are usually sparse within the candidate-site space, AUPRC is more sensitive to the enrichment of reference-positive sites among high-scoring candidates and was therefore interpreted together with AUROC as a major threshold-independent ranking metric.

For tasks with continuous references, tool-derived continuous scores were matched to external reference modification ratios by exact genome-level coordinates. Pearson correlation, Spearman correlation, coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE) were calculated to assess concordance between predicted and reference modification levels. Pearson correlation measured linear agreement, Spearman correlation measured rank-based agreement, and RMSE and MAE quantified numerical prediction errors. For heatmap visualization, RMSE and MAE were transformed into $1 - \text{RMSE}$ and $1 - \text{MAE}$ in selected figures, so that higher values consistently indicated better agreement across all displayed metrics.

G Downsampling-based Robustness Evaluation of m6A Detection

To evaluate the effect of reduced read input on the stability of m6A detection, we performed read-level downsampling analysis using the *Arabidopsis thaliana* VIRc dataset. This dataset was selected because its full-depth data volume and coverage were relatively high, allowing a sufficiently large candidate-site evaluation space to be retained even after reducing the read fraction. Reads were randomly sampled from the full FASTQ file to construct five read fractions: Full, 80%, 60%, 40%, and 20%. Reads in lower fractions were designed to be subsets of those in higher fractions, thereby reducing the influence of sampling randomness on trend interpretation.

Each read fraction was realigned to the same reference sequence to generate the corresponding BAM file. For tools relying on Nanopolish eventalign, indexing, eventalign, and subsequent tool-specific preprocessing and prediction were rerun for each fraction. For m6Anet, eventalign, dataprep, and inference were regenerated on each downsampled subset. For CHEUI, the required eventalign input was regenerated from the corresponding FASTQ/BAM files, followed by read-level and site-level prediction. For SWARM, the BLOW5 signal library preconverted from the full-depth FAST5 files was reused, rather than repeating BLOW5 conversion; signal features were re-extracted and predictions were rerun using the FASTQ/BAM files from each read fraction. For TandemMod, which directly scans FAST5 files, FAST5 symlink directories were constructed according to the read IDs in the FASTQ file of each fraction, and resquiggle, signal extraction, feature extraction, and model prediction were performed within each subset directory.

The downsampling analysis focused on reference-hit recall, recovered reference-hit count, full-depth TP-site retention, and positive-call composition. Reference-hit recall denotes the proportion of reference-positive sites recovered at a given read fraction. Recovered reference-hit count denotes the actual number of recovered reference-positive sites. Full-depth TP-site retention was used to assess whether TP sites detected under the full-depth condition were still retained after downsampling. Positive-call composition was used to compare the changes in the proportions of reference-supported calls and non-reference calls among positive calls.

TABLE S11. DOWNSAMPLING SETTINGS FOR THE A. THALIANA m6A ROBUSTNESS EVALUATION.

Read fraction	Mean depth	Median depth	Main purpose
Full	72.1x	12x	Baseline
80%	60.5x	11x	Mild read reduction
60%	48.5x	9x	Moderate read reduction
40%	35.8x	7x	Strong read reduction
20%	21.3x	4x	Low-input condition

H Analysis of computational resource

Except for inference steps officially documented to support GPU acceleration, all tasks were submitted and executed through the Slurm job scheduling system. The CPU server was equipped with two Intel(R) Xeon(R) Gold 6240 CPUs at 2.60 GHz, providing 32 CPU cores and 128 GB of memory, and running CentOS Linux release 7.5.1804. Each job was uniformly allocated 16 CPU cores. When a tool provided parameters for the number of threads or processes, these parameters were set to 16. For steps where the tool itself could not fully utilize multi-core resources, we retained the official or original workflow and did not introduce additional manual parallelization.

Steps officially supporting GPU acceleration were executed on a GPU node equipped with two Intel Xeon Gold 6330 CPUs, 56 physical cores/112 threads, approximately 503 GB of host memory, and one NVIDIA A100 GPU. For GPU jobs, runtime and host memory usage were recorded using `/usr/bin/time -v`, and GPU memory usage was recorded using `nvidia-smi`. Because some preprocessing steps involved extensive file I/O, runtime measurements may have been affected by I/O fluctuations on the shared server.

For large input files, CHEUI preprocessing was performed in parallel chunks. TandemMod was run in sequential batches during the signal extraction and feature extraction stages to reduce the data scale of individual jobs and facilitate rerunning failed batches. Runtime was calculated as the wall-clock time of the complete workflow, including splitting, preprocessing, prediction, and merging steps. Therefore, the runtime reported in this study reflects the computational cost under practical workflow execution rather than a strictly CPU-normalized comparison of algorithmic speed.

I Tool-specific Sensitivity and Upstream Optimization Analyses

In the above benchmark results, SWARM, as a newly developed tool, and m6Anet, as a classical m6A modification detection tool, both showed favorable performance. Therefore, we focused on these two tools for threshold and parameter analyses to explore the potential for further performance improvement.

To assess whether SWARM and m6Anet identified overlapping or complementary m6A candidate sites, we compared their fixed-threshold positive sets within the same common-site benchmark. SWARM-positive sites were defined as sites with probability ≥ 0.9972 and stoichiometry > 0.1 , whereas m6Anet-positive sites were defined as sites with probability_modified ≥ 0.9 . Based on these two positive sets, four strategies were constructed: SWARM-only calls, m6Anet-only calls, the intersection of the two tools, and the union of the two tools. Precision, recall, specificity, and F1 score were then calculated for each strategy to evaluate their potential utility in high-recall screening and high-confidence filtering scenarios.

To make SegPore outputs compatible with m6Anet input requirements, we first converted the SegPore Step 3 output into an m6Anet-compatible eventalign format and examined whether the event-level features, reference k-mer annotations, and coordinate conventions satisfied the input assumptions of m6Anet dataprep. When the original k-mer annotations in the SegPore-derived eventalign output were used directly, multiple reference_kmer values could occur at the same contig-position, which violated the m6Anet requirement that each coordinate should correspond to a unique reference_kmer. Therefore, we

reconstructed the reference_kmer, model_kmer, and their corresponding model mean and model standard deviation based on the GRCm39 reference genome. All subsequent analyses used the k-mer annotations reconstructed from the reference sequence to ensure consistent k-mer definitions at the same genomic coordinate.

In addition, we further recovered the true event-level features using the event standard deviation and event length information from the SegPore output, and recalculated the standardized level according to the reference k-mer model to reduce the influence of format conversion on m6Anet input features. Considering that SegPore-derived eventalign uses chromosome-level contigs, whereas the default readcount_max parameter in m6Anet is more suitable for transcript-level eventalign, we adjusted readcount_max to 100000 to avoid abnormal compression of the candidate space caused by default read-count truncation.

After completing full eventalign conversion and m6Anet inference, we further compared the prediction performance of the traditional m6Anet workflow and the SegPore-derived m6Anet workflow on the GGACT motif subset. Because the SegPore k-mer parameter table affects event alignment between signal segments and the reference sequence, we further performed iterative upstream parameter optimization for SegPore specifically on the GGACT motif. Specifically, the official SegPore 5-mer parameter table was used as the initial parameter set, and Step 3 alignment and Step 4 GMM parameter updating were alternately performed. In each iteration, event-level alignment was first generated based on the current parameter table. Then, a two-component Gaussian mixture model was fitted to GGACT-related events to estimate current distribution parameters, which were used to update the 5-mer parameters corresponding to GGACT. For each iteration, the mean, standard deviation, mixture weight of the two components, and the number of GGACT events were recorded to assess whether the parameters had converged.

After five rounds of GGACT parameter iteration, the corresponding SegPore-derived eventalign output was converted into an m6Anet-compatible input format and processed using the same m6Anet dataprep and inference workflow. Because m6Anet requires event information from regions surrounding candidate sites, GGACT-only eventalign was not used as the formal input. Instead, m6Anet inference was performed using the full eventalign output, after which GGACT motif sites were extracted from the site-level results. High-confidence positive sets and loose exclusion sets were then constructed within the shared GGACT candidate-site space to compare the ranking ability of prediction scores and the fixed-threshold performance between the traditional m6Anet workflow and the SegPore-derived m6Anet workflow. The evaluation metrics included AUROC, AUPRC, and fixed-threshold classification metrics.